CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: November 2019 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is Bytecode and Java Virtual Machine (JVM)?

Explanation: Bytecode is compiled intermediate instructions for the virtual JVM processor, enabling cross-platform portability.

Q2 (3M) List the four fundamental principles of Object-Oriented Programming.

OOP Principles: Encapsulation, Abstraction, Inheritance, and Polymorphism.

Q3 (3M) What is Garbage Collection in Java?

Explanation: Automatic reclamation of heap memory occupied by dereferenced objects.

Q4 (3M) What is Final variable, Final method, and Final class?

Final keyword: Variable becomes a constant, method cannot be overridden, class cannot be inherited (subclassed).

Q5 (3M) Define Constructor in Java.

Definition: A special method called automatically when an object is instantiated, having the same name as the class and no return type.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Interface in Java with code demonstrating multiple inheritance.

Interface: Contract with method declarations implemented by classes to achieve multiple inheritance.

Q7 (7.5M) Explain Applet Life Cycle and how Applets are embedded into HTML.

Applet in HTML: Embedded using `<applet code="MyApplet.class" width="300" height="200"></applet>`.

Q8 (7.5M) Explain Exception Handling using try, catch, and finally blocks.

Exception Handling: Structured handling preventing runtime crashes and releasing resources safely.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) What is JSP? Explain advantages of JSP and main components. How is JSP different from Servlet?

JSP vs Servlet: Comprehensive comparison, JSP compilation lifecycle, directives, actions, and scriptlets.

Q10 (15M) Explain Multithreading in Java: Thread Life Cycle, States, and Synchronization.

Thread Life Cycle: New → Runnable → Running → Blocked/Waiting → Terminated.

Q11 (15M) **Explain JDBC Architecture, Statement vs PreparedStatement, and write a program to insert student records.**

JDBC Program: Complete program inserting records using PreparedStatement to prevent SQL injection.

Q12 (15M) **Explain Event Handling in Java AWT/Swing with MouseListener and ActionListener.**

Event Handling: Source → Event Object → Listener interface implementation.

Q13 (15M) **Write HTML and CSS code to design a responsive institutional web page.**

Web Design: Clean HTML markup with external/internal CSS styling, tables, lists, and navigation bar.
